DATE: **EXPT NO:01**

EARTHQUAKE ANALYSIS SYSTEM USING MACHINE LEARNING

AIM

To analyze earthquake data using Exploratory Data Analysis (EDA), visualize important earthquake patterns, preprocess the dataset, build a Machine Learning model to predict tsunami occurrence, evaluate the model, and export the processed data to Excel.

PROCEDURE

1. Download the Earthquake dataset.
2. Import the required Python libraries such as Pandas, Matplotlib, Seaborn, and Scikit-learn.
3. Load the earthquake dataset using Pandas.
4. Display the first few records, dataset shape, information, and missing values.
5. Remove unnecessary columns such as title and location columns.
6. Check and handle missing or duplicate values.
7. Analyze important features such as magnitude, depth, latitude, longitude, cdi, mmi, sig, gap, and nst.
8. Visualize the distribution of tsunami occurrences using a count plot.
9. Create a visualization to identify the relationship between magnitude and tsunami occurrence.
10. Separate the input features (X) and target variable (y), where tsunami is the target.
11. Split the dataset into training and testing data.
12. Train a Machine Learning model using the training data.
13. Predict tsunami occurrence using the test data.
14. Evaluate the model using suitable performance metrics such as Accuracy Score and Confusion Matrix.
15. Export the processed earthquake data to an Excel file.
16. Display the analysis results.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# 1. Load dataset
df = pd.read_csv("earthquake_data.csv")

# 2. Display data
```

```
print(df.head())

# 3. Basic information
print(df.shape)
print(df.info())

# 4. Check missing values
print(df.isnull().sum())

# 5. Fill missing values
df["cdi"] = df["cdi"].fillna(df["cdi"].median())
df["mmi"] = df["mmi"].fillna(df["mmi"].median())

# 6. Remove unnecessary columns
df = df.drop(["title", "location"], axis=1)

# 7. Visualization - Tsunami
sns.countplot(x="tsunami", data=df)
plt.title("Tsunami Occurrence")
plt.show()

# 8. Visualization - Magnitude vs Tsunami
sns.countplot(x="magnitude", hue="tsunami", data=df)
plt.title("Magnitude vs Tsunami")
plt.xticks(rotation=20)
plt.show()

# 9. Convert categorical data into numbers
df = pd.get_dummies(df, drop_first=True)

# 10. Separate features and target
X = df.drop("tsunami", axis=1)
y = df["tsunami"]

# 11. Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 12. Train Logistic Regression model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# 13. Make prediction
y_pred = model.predict(X_test)

# 14. Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# 15. Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
```

```
sns.heatmap(cm, annot=True, fmt="d")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# 16. Export DataFrame to Excel
df.to_excel("earthquake_analysis.xlsx", index=False)
```

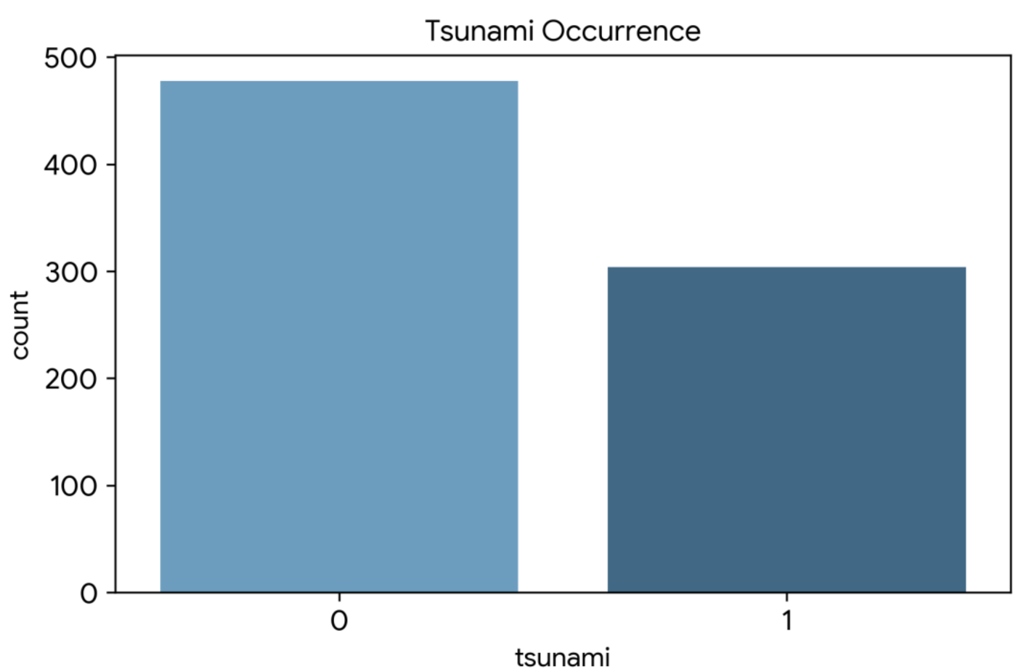
OUTPUT:

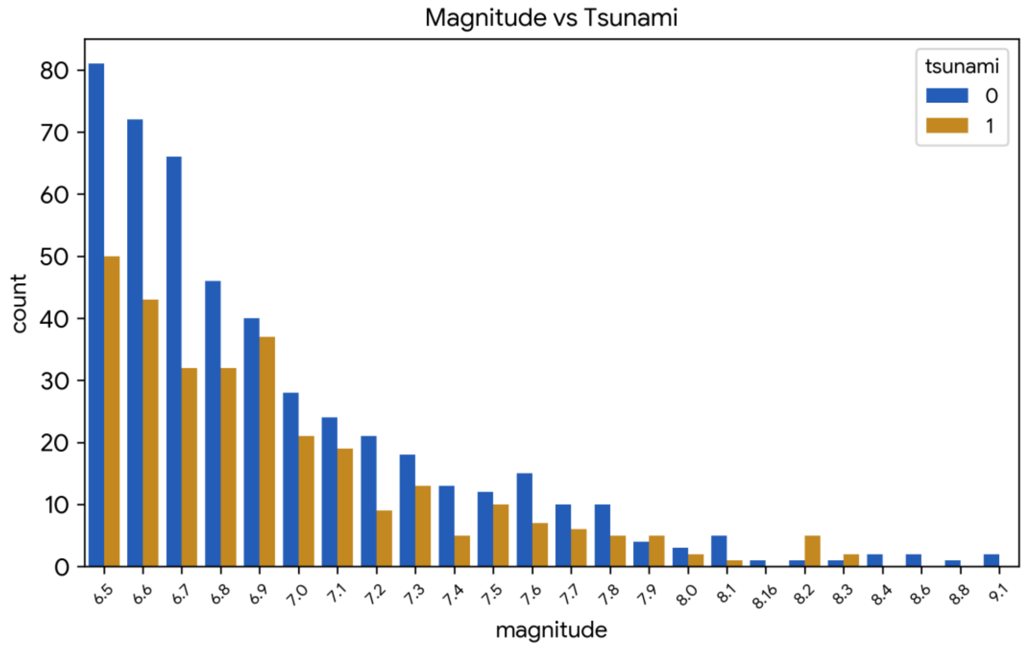
```
                                          title  magnitude         date_time  cdi
mmi  alert  tsunami  sig net  nst   dmin   gap magType    depth  latitude
longitude                  location continent         country
0  M 7.0 - 18 km SW of Malango, Solomon Islands        7.0  22-11-2022 02:03    8
7  green        1  768  us  117  0.509  17.0     mww   14.000   -9.7963    159.596
Malango, Solomon Islands   Oceania  Solomon Islands
1      M 6.9 - 204 km SW of Bengkulu, Indonesia        6.9  18-11-2022 13:37    4
4  green        0  735  us   99  2.229  34.0     mww   25.000   -4.9559    100.738
Bengkulu, Indonesia       NaN               NaN
2                                       M 7.0 -        7.0  12-11-2022 07:09    3
3  green        1  755  us  147  3.125  18.0     mww  579.000  -20.0508   -178.346
NaN   Oceania             Fiji
3           M 7.3 - 205 km ESE of Neiafu, Tonga        7.3  11-11-2022 10:48    5
5  green        1  833  us  149  1.865  21.0     mww   37.000  -19.2918   -172.129
Neiafu, Tonga       NaN               NaN
4                                       M 6.6 -        6.6  09-11-2022 10:14    0
2  green        1  670  us  131  4.998  27.0     mww  624.464  -25.5948    178.278
NaN       NaN              NaN

(782, 19)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 782 entries, 0 to 781
Data columns (total 19 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   title      782 non-null    object
 1   magnitude  782 non-null    float64
 2   date_time  782 non-null    object
 3   cdi        782 non-null    int64
 4   mmi        782 non-null    int64
 5   alert      415 non-null    object
 6   tsunami    782 non-null    int64
 7   sig        782 non-null    int64
 8   net        782 non-null    object
 9   nst        782 non-null    int64
 10  dmin       782 non-null    float64
 11  gap        782 non-null    float64
 12  magType    782 non-null    object
 13  depth      782 non-null    float64
 14  latitude   782 non-null    float64
 15  longitude  782 non-null    float64
```

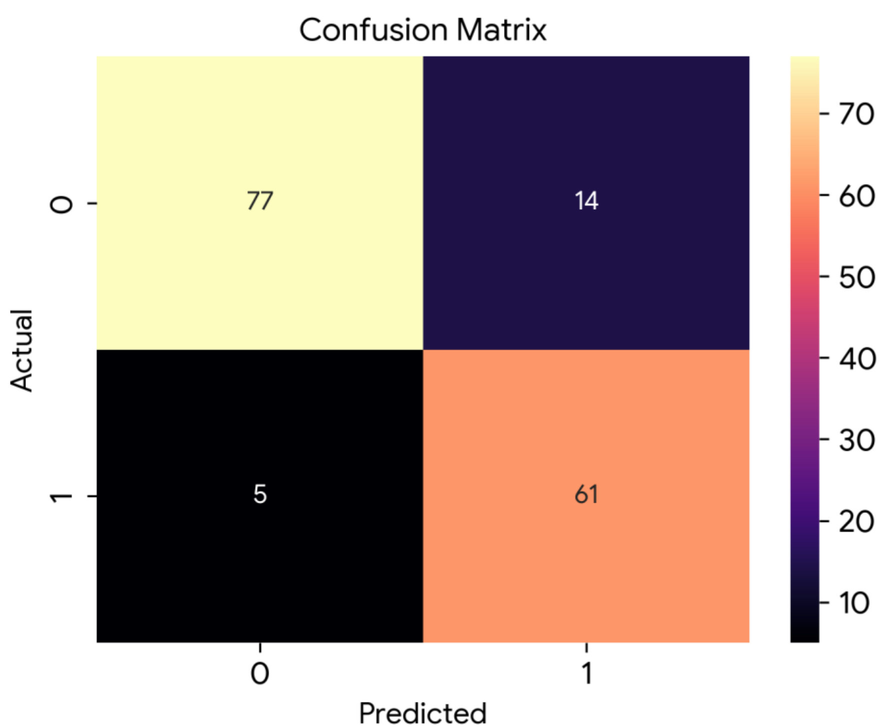
```
 16   location    777 non-null    object
 17   continent   206 non-null    object
 18   country     484 non-null    object
dtypes: float64(6), int64(5), object(8)
memory usage: 116.2+ KB
None

title          0
magnitude      0
date_time      0
cdi            0
mmi            0
alert        367
tsunami        0
sig            0
net            0
nst            0
dmin           0
gap            0
magType        0
depth          0
latitude       0
longitude      0
location       5
continent    576
country      298
dtype: int64
```






```
/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/_logistic.py:465:
ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.
Accuracy: 0.8726114649681529
```



RESULT:

The earthquake dataset was successfully imported, analyzed, visualized, and processed. A Logistic Regression model was trained to predict tsunami occurrence, and the model performance was evaluated using accuracy and a confusion matrix. The processed data was also exported to an Excel file.